Aayush Raj

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SUMMARY

Computer Science undergraduate with a strong foundation in Data Structures And Algorithms, Object-Oriented Programming, Operating Systems, and DBMS. Proficient in C/C++, Java, and Python, with hands-on experience building software applications using Flask, REST APIs, OpenCV, and SQL. Strong problem-solving and debugging skills with experience in performance optimization, software development, and deployment. Eager to apply strong programming and analytical skills to challenging engineering problems.

Technical Skills

Languages: Python, C++, java, SQL
Backend: FastAPI, Flask, REST APIs
Libraries: Pandas, NumPy, Matplotlib, Seaborn
Databases: MongoDB, MySQL
Fundamentals: Data Structures And Algorithms, DBMS, Operating Systems, Computer Networks
Tools/Technologies: Git, GitHub, Docker, Swagger/OpenAPI
Cloud (Basics): AWS (EC2, S3), CI/CD Fundamentals.

Education

•**Jain University, Bangalore** *2023-2027*
Bachelor of Technology in Computer Science(Software Engineering)

Projects

•**Weather Web Application (Flask)** *Interactive Web Application* Source Code

- Built a full-stack weather application using Flask (Python) and integrated the OpenWeatherMap API to retrieve real-time weather data and 5-day forecasts.
- Developed RESTful APIs and documented them using Swagger UI/OpenAPI, improving API testing and maintainability.
- Implemented features such as geolocation-based weather search, temperature unit conversion (Celsius/Fahrenheit), and display of humidity, wind speed, and weather conditions.
- Designed a responsive UI with light/dark mode, dynamic weather icons, and loading feedback, enhancing user experience across devices.

•**SmartAttend – Face Recognition Attendance System** *Python, Flask, OpenCV, MongoDB Atlas, JavaScript* Source Code

- Built SmartAttend, a Flask-based attendance management system with secure admin authentication, student profiles, attendance tracking, and CSV/Excel reporting..
- Integrated MongoDB Atlas using MongoEngine, with database-level rules that prevent duplicate attendance entries for the same student and day.
- Developed an OpenCV-powered facial enrolment and recognition workflow that captures face profiles and marks attendance with configurable confidence thresholds.
- Deployed the application on Render with environment-based secret management, persistent biometric-file storage, and responsive web dashboards..

Certifications

•**Python Programming** IITK Foundation for Innovation

•**Alpha (DSA with Java)** Apna College

•**Introduction to Web APIs** LinkedIn Learning